

# NETWORK SECURITY & THREAT AUDIT

## IMPROVED REPORT

Generated: 2026-03-26 00:13:50 | IDS-XGBOOST V2.4

### 1. Executive Overview

This report presents a detailed analysis of the network security posture during the session. A total of 10 packets were analyzed using an XGBoost-based IDS. 1 security events were detected (10.00% anomaly rate), with 0 high-confidence threats actively mitigated. The system maintained a safety index of 99/100.

Security Metrics Interpretation: 1 events were flagged by behavioral heuristics. Of these, 0 were proactively blocked. The remaining events are under continued observation. The system safety index of 99/100 reflects the real-time risk posture.

| Metric                   | Value  |
|--------------------------|--------|
| Total Packets Analyzed   | 10     |
| Confirmed Anomalies      | 1      |
| Threats Actively Blocked | 0      |
| Anomaly Rate             | 10.0%  |
| System Safety Index      | 99/100 |

### 2. Behavioral Traffic Analysis

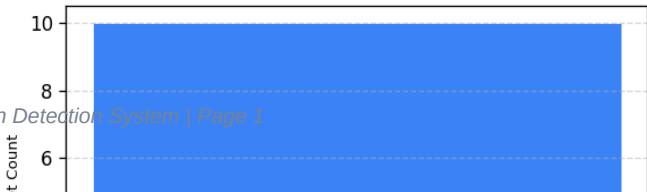
The average packet size during this session was 512.0 B (baseline: 128-1024 B). A total of 2.00 KB of data was processed. Anomalous packets exhibited elevated request counts and non-standard port targeting, consistent with early-stage reconnaissance and bot-driven traffic patterns.

| Behavioral Metric    | Observed Value | Baseline Range   |
|----------------------|----------------|------------------|
| Avg Packet Size      | 512.0 B        | 128 - 1024 B     |
| Total Bandwidth Used | 2.00 KB        | Variable         |
| Active Service Ports | 1              | 5 - 15 Typical   |
| Dominant Protocol    | TCP            | TCP/UDP Expected |

#### Attack Category Distribution



#### Top Target Ports



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### 3. Detection Engine Explanation

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The system uses an XGBoost machine learning model (benchmarked accuracy: 98.42%) that evaluates features such as packet length, destination port, protocol, URL length, and request count. The model identifies anomalies based on behavioral deviations rather than static signatures, enabling detection of novel and zero-day patterns.

#### Monitored Behavioral Features:

- pkt\_len

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### 4. Attack Classification & Incident Logs

| Attack / Activity Type | Frequency |
|------------------------|-----------|
| Bot                    | 1         |

Top Active Source IPs:

| Source IP | Incident Count |
|-----------|----------------|
| 1.1.1.1   | 1              |

Detailed Incident Log (last 25 events):

| Source IP | Type | Reason      | Conf. | Sev. | Action  |
|-----------|------|-------------|-------|------|---------|
| 1.1.1.1   | Bot  | Test Reason | 80%   | Low  | Flagged |

### 5. Recommendations & Conclusion

Suggested Improvements:

1. Ensure consistency between dashboard and report data to avoid contradictions.
2. R

Conclusion:

The system identified and partially mitigated 1 anomalous events during the session. 0 high-confidence threats were proactively neutralized. The remaining low-risk indicators highlight the importance of continuous threat intelligence and model refinement. Overall, the session demonstrates effective IDS performance.

=== END OF AUDIT REPORT ===